

**Viswa Group Proposal**  
**Dated: 26<sup>TH</sup> April 2021**

## **EEXI and CII Compliance** **(A Calculation, Implementation and Improvement Project)**



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# Introduction

With the Greenhouse Gas Strategy towards 2050, the IMO has set the goal to reduce carbon intensity by 40% within the next decade up to 2030 and by 50% in total (70% intensity) up to 2050. The Greenhouse Gas Strategy was approved by the IMO in 2018. The reduction rates are related to the baseline of 2008. Short-term, mid-term and long-term measures are distinguished to achieve the goal.

Short-term measures are meant to be set into force by 2023. Different measures can be considered and implemented, for example the introduction of speed limits, and innovative measures such as air lubrication to reduce hull friction etc.

With the proposed implementation of EEXI and CII factors by IMO, it becomes imperative to confirm compliance of each vessel in a fleet to meet the requirements.

The consultancy services highlighted in this document will provide all necessary calculation and processes to meet the business goals for regulatory compliance.

Client will provide certain data such as IMO DCS and voyage basis data towards calculations of Annual Emissions Ratio and EEOI.

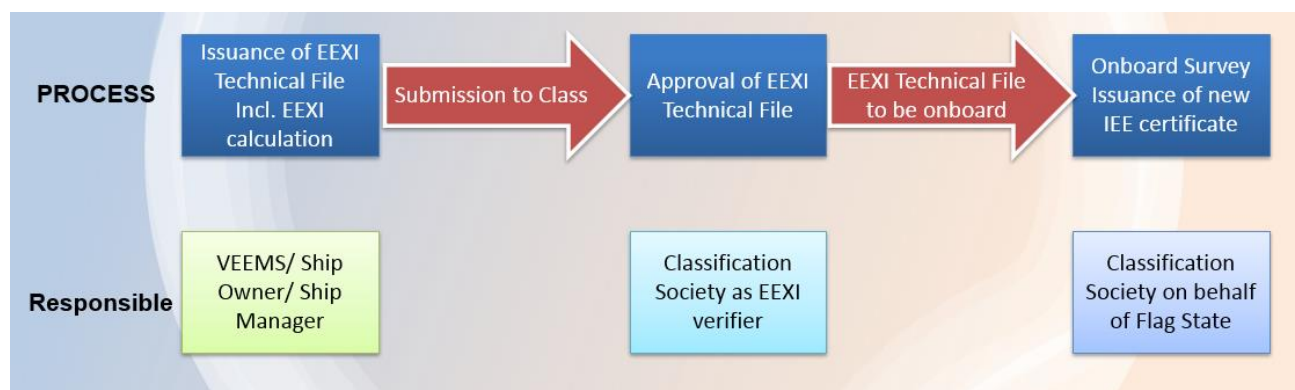
Insight on how real-time data can be monitored live towards studying the implementation of innovative measures towards continuous monitoring of vessel's EEXI maintenance through the Viswa Group's VEEMS (Vessel Energy Efficiency Management System) software and can be clubbed together with this scope with future discussion.

## Work scope

### 1. EEXI Analysis

#### a. Current EEXI

The draft IMO guidelines for the calculation of EEXI are yet to be formally adopted. The EEXI will be calculated for every ship where attained EEDI is non-compliant with the required EEXI. EEXI will be calculated based on the best available information.



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A list of the information required for each ship to be assessed is as follows:

- I. Trim & Stability booklet
- II. Ship specific sea trials report (containing scantling and ballast draught Speed-Power details).
- III. Ship specific shop test reports for Main and Auxiliary Engines
- IV. General Arrangement and Capacity Plan
- V. Ship specific signed sea trials report (containing scantling and ballast draught power curves)
- VI. Ship specific model report (containing scantling and ballast draught power curves)
- VII. Shaft power take-off (PTO) or -in (PTI) installed unit details.
- VIII. Usage of any energy saving devices (ESD's) or Propulsion Improving Devices (PIDs) installed on each specific ship

Where the information required is not available logical assumptions will be derived and used to calculate the EEXI values.

#### **b. Required EEXI**

Required EEXI will be calculated for each ship as per the vessel type and size basis MARPOL Annex VI, Regulation 19-21.

The attained EEDI should be less than the required EEXI.

For ships where attained EEDI is already less than required EEXI, the client can seek direct approval by Class.

#### **c. EEXI improvement for non-compliant ships**

For ships that are identified to be non-compliant with the EEXI limits as per Sections 1 and 2 mentioned above, various options can be adopted to reach compliance as explained below.

##### **I. Retrofitting with Propulsion Improvement devices (PID)**

There are various technologies such as PBCF, Mewis duct, air lubrication, Flettner rotors, etc. available in the market. A detailed technical study for a target ship will be carried out to accurately determine its effectiveness in EEXI improvement.

CFD or model testing will be studied to gauge efficiency.

VEEMS will be provided to capture live data to arrive at calculations presenting the actual improvement achieved by the subject vessel.

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The cost of such products/ services available in the market will be evaluated.

The fee for this study will be evaluated with the client with a different proposal outside of this scope document.

## **II. Shaft or Engine Power Limitation**

The percentage shaft or engine power limitation (SHaPoLi/EPL) required to achieve compliance will be calculated. This will however also result in the reduction of the vessel's maximum speed which will be estimated. If this is the preferred option, the engine manufacture will need to be contacted for carrying out this work at the earliest possible location. Subject vessel's Class will be required to witness the installation of the SHaPoLi/ EPL.

A minimum power assessment will be carried out as per ships that are subject to MARPOL Annex VI Regulation 21 requirements for minimum power as per MEPC.1/Circ.850/Rev.2.

Engine manufacturer's work and Class attendance are outside this document's scope.

## **III. Shaft Power or Engine Power Limitation management plan**

A management plan will need to be prepared if (SHaPoLi/EPL) is decided to be used for EEXI compliance. This document will be subject to verification by the Administration. PSC can also refer to the management plan during inspections.

The Viswa Group will provide this management plan including the relevant supporting document from the engine manufacturer covering the areas of the technical description of the system and sealing arrangement adopted onboard, inspections and procedures to confirm system condition, and procedures and logs related to this system's release.

## **IV. EEXI technical file**

The Technical file will be prepared including the details that were performed for EEXI calculations. The EEXI Technical file will be prepared according to the draft EEXI guidelines. Any additional information such as survey reports of SHaPoLi/ EPL etc. will be included. This file will be submitted to Class for approval and associated Class fees are outside of this scope.

## **2. Carbon Intensity Indicators**

There are two indicators for calculating operational Carbon Intensity:

- a. Annual Efficiency Ratio (AER)  
IMO DCS Annual Emissions reports for all the vessels will be reviewed to obtain the following information for each ship:

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- ✓ The start and end date of the assessment period
- ✓ Miles sailed.
- ✓ Fuel consumption for each type of fuel consumed .

The Statement of Compliance (SoC) will be used to confirm that the data contained in the emissions report has been verified by an approved recognised organisation.

The verified data will then be used to carry out the following calculations for each ship in the fleet:

- ✓ Total carbon dioxide emissions based on the carbon factors listed in the IMO document.
- ✓ The annual efficiency ratio (AER)
- ✓ The trajectories published by the Poseidon Principles Association will be used to calculate the percentage difference for a ship's AER for each ship as well as the year it will become misaligned if the AERs do not change (note it is likely that the IMO will adopt a very similar AER trajectory in the future)

b. Energy Efficiency Operational Index (EEOI)

- ✓ EEOI calculations will be performed basis on the voyage-wise data supplied.
- ✓ Calculations using the climate alignment of all the vessels can be used to make informed business decisions to achieve future CO2 reduction targets.

### 3. Documents required

- a. Statement of Compliance (SoC) and Verified IMO DCS Annual Emissions Report for 2020 – For each ship.
- b. Datasheet containing every voyage detail that each vessel has done including at least Cargo carried, distance sailed (NM), and fuel consumed for 2020 (This sheet will be like EU MRV reporting which is a voyage-by-voyage data aggregation for EU voyages but applied for non-EU voyages as well).
- c. Fleet list containing IMO number, GT, installed Main Engine Power, and DWT.
- d. Other items mentioned in section (1.a) earlier.
- e. If SHaPoLi/ EPL is implemented below details will be required:
  - i. Work report produced by the manufacturer, including the description of the Main Engine EPL device or SHaPoLi system installation report, method of sealing or locking and procedure for releasing the EPL, and/or method of limiting the power and procedure for releasing the SHaPoLi.
  - ii. Image evidence of any seals and or limiters.

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**4. Delivery Timeline (per ship)**

Assuming all the requested documents are available, the timeline per ship is as below:

- a. EEXI calculation – 5 days
- b. AER and EEOI calculation – 2 weeks

**5. Service Fees (Per vessel)**

- a. EEXI Initial calculation – USD 445
- b. EEXI Technical file preparation – USD 445
- c. EEOI calculation – USD 445
- d. AER calculation – USD 395
- e. Sea Trial attendance and Report – USD 6495

For ships identified as EEXI non-compliant below rates apply:

- a. EEXI improvement calculation – USD 235
- b. SHaPoLi/ EPL Management Plan – USD 1395
- c. CFD (Computational Fluid Dynamics) - Feasibility study of PIDs (Propulsion Improvement Devices) – Quoted price will depend on the type of device(s)

**6. References**

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